

Regular Articles

From 5G to beyond: Passive optical network and multi-access edge computing integration for latency-sensitive applications

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ABSTRACT

A recent paradigm shift in support of 5G-and-beyond (5GB), Human-to-Machine/Robot (H2M/R), and the Tactile Internet has resulted in a surge of latency-sensitive applications being delivered across communication networks. These applications in conjunction with exponential growth in connected devices, have pushed for architectural and capacity changes in both wired and wireless networks. Passive optical networks (PONs) have evolved in terms of their capacity and function over the years and are now supporting both fixed as well as x-haul solutions to deliver cost-efficient, low-latency and high-bandwidth applications to end-users. In combination with multi-access edge computing (MEC), high computational tasks of applications can be offloaded to servers much closer to end-users further reducing the network latency. To this end, this paper presents a comprehensive review of the capabilities of integrated PON-MECs in delivering emerging latency-sensitive applications. Specific use case scenarios benefiting from integrated PON-MECs are also detailed. Shared learning through a federated PON-MEC framework to expedite resource allocation decisions is discussed and finally, the time-savings arising from implementing the proposed framework is quantified, showing further improvements in network latency performance.

1. Introduction

With the proliferation of 5G networks, Internet-of-Things (IoT) [1], and Machine-to-Machine communication (M2M) [2], attention is now focused on next-generation network paradigms including 6G networks and the Tactile Internet [3,4]. Wired and wireless communication networks will continue to rapidly evolve both in terms of their architectures and capabilities to support the surge in latency-sensitive human-to-machine/robot (H2M/R) applications such as augmented reality/virtual reality (AR/VR) [5], industry automation [6], tele-surgery [7], and intelligent transport systems (ITS) [8]. The expedited growth in the number of mobile devices, projected to reach 29.3B in 2023 and with each envisioned to generate large amounts of data, has also warranted new network architectures, hardware, and software solutions [5,9,10].

Today, the seamless delivery of latency-sensitive and reliable services in 5G networks is realized by enabling technologies such as multi-access edge computing (MEC) [9,11] and fog computing [12]. Prior to MEC, cloud services such as computing, storage, and caching were performed at centralized data centers, often at a long distance

away from end-users [13]. As a result, user data experienced high propagation delays that prevented the underlying networks from meeting stringent end-to-end latency requirements of applications. To overcome this problem, MEC enabled cloud computing and IT services to be brought closer to the end-users through the placement of MEC servers at the access network edge [9,14,15]. Rather than having the data traverse through the mobile backhaul or core networks, computing, storage, and caching are now rendered at the access network edge [16].

Recent studies on MEC and 5G have highlighted the benefits of using optical fiber networks to support the increasing number of connected devices and their quality-of-service (QoS)-aware applications. Optical fiber benefits from high capacity and low propagation delay. Taking advantage of optical fiber that is already deployed in a majority of urban areas close to residential and industrial premises, the cost-efficient and high-capacity passive optical network (PON) has become a natural choice for fronthauling and backhauling of 5G and 5GB networks [17]. The PON has evolved over the years in terms of its architecture, hardware, capacity, and underlying protocols and algorithms [18]. These developments in conjunction with optical fiber deployments that

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have now reached residential and business premises, have enabled the delivery of bandwidth-hungry and latency-sensitive services to large numbers of users.

The integration of PON and MEC architectures is a potential solution to deliver latency-sensitive yet task-intensive H2M/R applications. Such applications define the capabilities of 5GB and 6G as they require latency in the milliseconds range, a ultra-reliable network connectivity and a significant amount of data processing. In order to satisfy such stringent QoS requirements, the suitability of the converged PON-MEC network from different aspects such as PON and MEC configurations, resource allocation across heterogeneous network segments, and complementary technologies such as machine learning that can further reduce end-to-end latency through added intelligence at the mobile edge, should be carefully explored. Hence, in this paper, we review PON, MEC and PON-MEC integration in detail and investigate how their inherent characteristics assist low-latency and task-intensive applications. To demonstrate the capabilities of integrated PON-MECs in supporting latency-sensitive applications, we propose a bandwidth allocation decision framework that uses reinforcement learning to reduce bandwidth allocation decision time and sharing this learning across a multiple PON-MECs that form a federation to further improve latency performance.

The remainder of this paper is structured as follows. In Section 2, we present a review of PON and MEC architectures. In Section 3, we discuss the PON-MEC integration architectures and specific use case scenarios of emerging latency-sensitive applications that would benefit from an integrated PON-MEC network. In Section 4, we describe our framework that expedites bandwidth decision making in federated PON-MEC for H2M/R communications. Finally, the main contributions of this paper are summarized in Section 5.

2. Background: Passive optical networks and multi access edge computing

In this section, we provide an overview of major PON and MEC architectures and their inception to recent deployments, which serves as the foundation to discuss the main concepts covered in the rest of the paper.

2.1. PON overview

Over the last decade, we have witnessed the unrelentless penetration of optical fiber deeper and deeper into residential and business premises. According to [19], 40 percent of the world’s population will have fixed Fiber-To-The-x (FTTx) connection by 2026. The models of FTTx, namely FTTH (home), FTTC (curb) and FTTB (building), installed either through a physical point-to-multipoint (P2MP) or point-to-point (P2P) topology offer direct fiber connection to or close to the end-users. The global FTTx market is still growing with a steep trajectory, predicted to reach 22.4B by 2026 [20].

Most FTTx networks are based on the PON technology due to its cost-effectiveness and low energy consumption per bit. The PON has a P2MP topology with an unpowered optical distribution network (ODN). Specifically, the ODN comprises a feeder fiber that connects the optical line terminal (OLT) at the central office (CO) to a passive splitter and a number of distribution fibers that connects the passive splitter to optical network units (ONUs) at end-user locations. For N subscribers, a PON requires only the installation of a single transceiver at CO and N transceivers at the ONUs, giving rise to $N+1$ transceivers overall. The shared feeder fiber saves CO space and allows for easy termination. This physical topology also allows its downstream and upstream bandwidth to be naturally shared through time division multiplexing (TDM) and time division multiple access (TDMA), respectively. The following are some of the TDM/TDMA-based PON architectures standardized in recent literature.

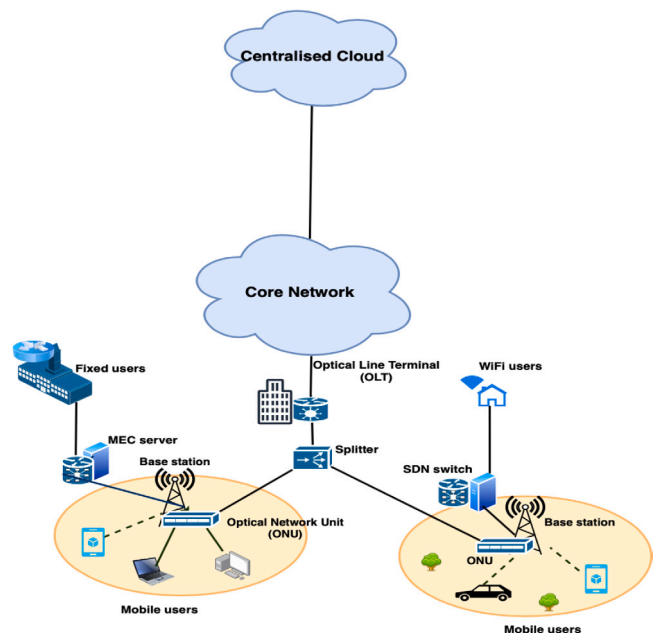


Fig. 1. Multi-Access Edge Computing (MEC) architecture [33,34].

- GPON and 1G-EPON
 - ITU-T G.984 Gigabit PON (GPON) [21]
 - IEEE 802.3ah Gigabit Ethernet PON (1G-EPON) [22]
- 10-Gigabit-Capable PON
 - IEEE Std. 802.3av 10 GE-PON [23]
 - ITU-T G.987 10G-PON standard [24]
 - ITU-T G.9807.1 XGS-PON [25]
- NG-PON2
 - ITU-T G.989 Next-Generation PON 2 (NG-PON2) standard [26]
- 50-Gigabit-Capable PON
 - IEEE 802.3ca 25G/50G-EPON standard [27]
 - ITU-T G.9804 High Speed PON (HSP) [28]

2.2. MEC overview

The concept of MEC evolved from mobile cloud computing (MCC) where mobile devices accessed remote centralized data centers through the mobile network to offload their computation and data processing [29]. The downside of MCC is the user data having to traverse through the core and access networks to reach centralized cloud resources, thereby experiencing significant propagation delay and congesting the backhaul and core networks [30]. As the number of mobile devices and the QoS requirements of their applications continued to grow, in order to enjoy the benefits of MCC while also meeting latency and other QoS requirements, cloud and IT services had to be brought closer to the end-users. For this purpose, Mobile Edge Computing which is an initiative of the ETSI Industry Specification Group (ISG), was identified as a favorable solution to integrate edge computation into mobile networks [9,31,32].

Fig. 1 illustrates the placement of mobile edge computing servers at a close proximity to end-users in a radio access network (RAN). Specifically, the servers are placed at multiple locations including the LTE macro base station, radio network controller (RNC) or the multi-radio access technology (RAT) cell aggregation point [31]. The servers

function to either process requests from end users in its vicinity or forward requests onto the centralized data centers for further processing. Realizing the possibility of serving both mobile and wired networks through edge computing in order to deliver latency-sensitive and computation-heavy applications, ETSI further redefined Mobile Edge Computing as Multi-Access Edge Computing (MEC) in which the new MEC architecture is defined as one that serves both wired and wireless networks at the edge [35,36], as shown in Fig. 1.

3. PON-MEC integration

3.1. Network architecture

It has emerged in recent literature the integration of both PON and MEC as a favorable solution in provisioning broadband access to meet QoS requirements. In [13], the authors investigate how to optimally plan a WDM-PON-based MEC-enabled fiber wireless access network (MFWAN) for latency-sensitive applications. The proposed work presents a mathematical model to minimize the end-to-end latency using constraints such as maximal transmission distance, maximal PON power budget, and bandwidth requests. In [37], the authors investigate different optical fronthaul architectures to support ultra-low latency and reliability required in 5G mobile fronthaul. Taking Tactile Internet as a use case, the authors of [3] present a decentralized cooperative dynamic bandwidth allocation algorithms for end-to-end resource coordination in a MEC-enabled fiber-wireless (FiWi) access networks using an EPON. Additionally, the authors of [38] study the collaborative computation offloading in fiber-wireless networks, specifically considering the resource allocation between a mobile device and the MEC server, and also between the mobile device and the centralized server. The authors of [33] also propose a dynamic resource allocation algorithm considering both centralized cloud and MEC enabled fiber-wireless networks.

The architecture of an integrated PON-MEC network is schematically shown in Fig. 1, whereby the OLT of the PON interconnects a centralized cloud (via the core network and the Internet) with N number of ONUs (via a passive splitter). The ONUs co-exist with mobile base stations (BSs) and are connected to MEC servers placed at the edge of the RAN. Under such an integrated PON-MEC network, the MEC servers support not just the mobile users but also wired customers through the ONUs. It is also important to note that not all mobile services require mobility and therefore would benefit from the fixed optical network resources connected to the RAN.

3.2. Use cases

As discussed in the previous section, both PON and MEC have inherent characteristics to support QoS-aware emerging use cases. In this section, insights into how the integration of high-capacity PONs with MEC capability can address QoS requirements in some use case scenarios are highlighted.

3.2.1. X-hauling 5GB wireless networks

Driven by the increasing number of mobile devices and the use of smart mobile applications, mobile technology is continuously and rapidly advancing. The rapid deployment of 5G wireless technology has reshaped network capabilities in an unimaginable way. To address the ongoing issue of providing mobile connectivity to an ever growing number of connected devices and with increasingly stringent QoS requirements, both academia and industry have recently begun to research the sixth generation (6G) mobile technology. According to [39], 6G mobile technology is envisioned to provide a peak data rate of 100 Gb/s, a user-experienced data rate of 1 Gb/s and an ultra-low latency of less than 1 ms over ubiquitous 3D coverage areas.

These high capacity future 5G/6G wireless networks must be supported by complementary high-capacity and secured transport networks for backhauling/fronthauling (x-hauling) the wireless cells to

transport all wireless data to the metro or core networks [40]. In particular, emerging 6G wireless networks is envisioned to consist of ultra-small wireless cells to achieve the required capacity and coverage requirements. As higher cell densities can lead to costlier x-haul transport connectivity [41], such cell densification calls for cost-efficient x-haul solutions in order to meet global business goals.

A traditional transport network typically consists of direct P2P fiber/wireless links between cell sites and the network core, which is commonly referred to as backhaul. However, to cope with the increasing demands in a cost and energy-efficient manner, a new paradigm of centralized processing of mobile signals was first introduced with 4G technology [43]. Cloud/centralized radio access network (C-RAN) is one of the main architectures currently deployed as centralized processing allows for advance wireless cooperative communication functionalities. In C-RAN, all functions are centralized in a baseband processing unit (BBU) which is placed at a central location. Radio frequency (RF) functionalities are performed at the remote radio head (RRH), located at the antenna cell site. In this architecture, RF signals are digitized and sent over fiber to the centralized BBU using Common Public Radio Interface (CPRI).

In a native C-RAN, the bandwidth required for this digitized signal scales with the physical parameters of antenna, e.g., antenna port count and wireless bandwidth in use. However, with the high number of antenna ports in use in 5G and beyond networks, the fronthaul between the BBU and RRH consequently demands exceedingly high-capacity links. As an example, using CPRI to support a fronthaul with 200 MHz of bandwidth, 64 QAM, 8X12 MIMO, and 96 antenna ports, will necessitate a 896 Gb/s link [44]. The demand for such high-capacity links makes C-RAN an impractical architecture.

A solution to this problem is to move some functionalities back to the cell site. There are ten different standardized ways, termed functional split options, in which functions can be split between the BBU and RRH [45]. In the latest 5G New Radio (NR) transport architecture, BBU functions are split into three entities. The centralized unit (CU), distributed unit (DU), and remote unit (RU). This new design in 5G facilitates RAN virtualization with flexible assignment of computing resources. The functional split options and their required x-haul data rates are illustrated in Fig. 2.

In comparison to other competing access technologies, PON can play a critical role in realizing high-capacity yet low-cost x-haul links [46]. To date, different PON architectures have been investigated for this purpose [46–48]. As an example, ITU-T G.Supp66 investigated traditional and emerging PON technologies to support different functional split options, while considering different bandwidth, latency, and other x-haul and 5G parameter constraints [48]. It is also worth mentioning that while the PON is unable to support low layer functional split options such as Option 8 and 7.1 due to its capacity limitation, it is still a suitable candidate to support functional split options above 7.2 (refer to Fig. 2). Specifically, PON can easily support 5G transport with the 7.2 or higher layer functional split options, and this specific use of PON is considered in the latest RAN architectures such as Open Intelligent RAN (O-RAN) [49]. With the principles of intelligence and openness in mind, the O-RAN architecture is standardized for building a virtualized RAN on open hardware with the help of artificial intelligence assisted radio control.

There are however existing challenges that requires addressing if considering PON-based x-haul solutions. Depending on the actual functional split option used, a key issue concerns the realization a scalable and cost-effective deployments. Several studies have investigated approaches to plan and dimension PON-based x-haul cost-effectively [42,50,51]. For example in [42], the authors propose a generalized integer linear programming based optimization framework that can be used to jointly plan both PON-based x-haul and 5G wireless networks to provide the cost-optimal network design in any given deployment scenario and irrespective of the RAN architecture used. Further, the

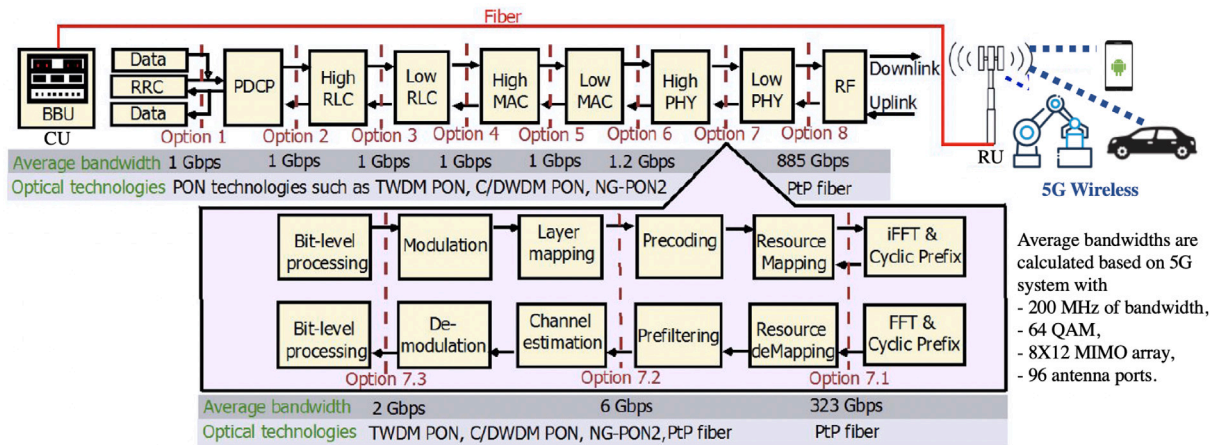


Fig. 2. 5G functional split options with supporting optical x-haul technologies [42].

framework takes into consideration a diverse set of network requirements such as network coverage and end-user capacities. Analyses on different 5G RAN and x-haul deployment scenarios show that PON-based x-haul options can save more than 30% to 40% of the deployment cost compared to P2P fiber-based x-haul option.

Another key issue revolves around resource management and allocation. In both PON and 5G networks, all RUs and end-users respectively, share the same bandwidth. Due to this shared bandwidth, two major factors can severely affect the latency, namely the discovery window and the bandwidth allocation algorithm. The discovery window, which is periodically opened by the OLT for newly registering ONUs, can cause delays in the x-haul. Different studies have looked at minimizing the impact of discovery window on network latency. For example, in [52], the authors propose a mechanism to assign the discovery window by using network slicing in order to reduce network latency. Effective approaches to reduce latency using intelligent bandwidth allocation algorithms have also been explored in [52–56]. In [52], the authors propose a cooperative bandwidth allocation by considering the actual data arrival period of fronthaul traffic to improve the latency and the bandwidth utilization. Further, in [33] the authors investigate the bandwidth allocation of an integrated PON-MEC x-haul and correspondingly propose a dynamic resource allocation algorithm to meet network and user requirements.

Looking further into the near future, integrated PON-MEC x-haul for 5GB networks, should include complementary technologies including machine learning (ML), network slicing, and NVF in order to meet the stringent service requirements. Further research into architectures, resource allocation mechanisms and a design framework is warranted in order to achieve full convergence between optical and wireless access technologies, ML, network slicing and MEC.

3.2.2. e-Health

Healthcare has evolved significantly over the years. Services that previously required in-person interaction with healthcare professionals are now being delivered remotely. For example, intelligent health surveillance systems have been proposed using wearable IoT sensors, which collect data pertaining to user behaviors. This sensor data is then processed and analyzed using ML and deep learning (DL) techniques to identify user patterns and anomalies. Such applications however, are latency sensitive and require significant processing resources which an end user may not be equipped with [60]. As such, MEC has been proposed for such applications to offload their computation and processing tasks to an edge server placed close to the end-user. Another emerging e-health application gaining interest is remote surgery, where a skilled surgeon performs a surgery on a patient who is remotely located. The surgeon controls a surgical robot placed at the patient side who mirrors

the actions taken by the surgeon. Due to the life-critical nature of this application, the end-to-end latency is expected to be less than 1 ms with a reliability of 99.999% [36]. Applications of this nature will benefit from PON-MEC integration as the PON will provide high bandwidth and reliability while the MEC ensures faster processing of data using ML and DL at the network edge.

Fig. 3 depicts integrated PON-MECs supporting different use case scenarios. More specifically, MEC servers are placed at the edge of the network and are connected to the ONUs placed at the base stations. For health surveillance applications, data generated by the user wearable sensors will be analyzed in these MEC servers in order to identify user patterns and to send warnings or notifications to carers or emergency services in case of an emergency. The location and context-awareness together with high-capacity optical fiber connections will ensure latency-sensitive service delivery in this scenario. In applications such as remote surgery, the MEC-PON integration itself may not be able to satisfy stringent latency requirements. Additional techniques, including ML to predict the resource requirement of the end-users, are also necessary [61]. Both MEC and centralized servers can be used for this purpose, depending on their computation capabilities and trained data stored at the MEC for reduced latency.

3.2.3. Industry 4.0

Industry 4.0 involves improving the operation of traditional factories such that they are able to run more efficiently and adapt quickly to changing requirements of the production lines [62]. A shift in this direction is made possible by novel IoT technologies, data analysis techniques, and network improvements such as edge computing. As discussed in [63], a large amount of data is being generated at factory settings. These data can be analyzed and be converted to intelligence which can be used to optimize different aspects of the manufacturing process [64]. For example, ML-based predictive maintenance can be used, where a failure of a component/device is predicted beforehand to minimize the downtime of the manufacturing process. The human-agent-robot teamwork (HART) is another promising concept that can be used in factory floors to provide either location-dependent or location-independent but computation heavy services [58]. The process can be optimized to select either a resource-limited robot for tasks that requires moving from one place to the other or a resource-rich cloud agent for tasks that requires heavy computation.

The factory automation services of this nature deals with controlling the equipment in real-time, which would need latency in the range of 25 μ to 10 ms depending on the application and significant data processing. Further, due to the critical nature of these applications, a reliability with packet loss rate no less than 10^{-9} is expected [65]. As such, a network architecture that combines the low-latency computation of MEC and reliability and high bandwidth of PON is preferred. The

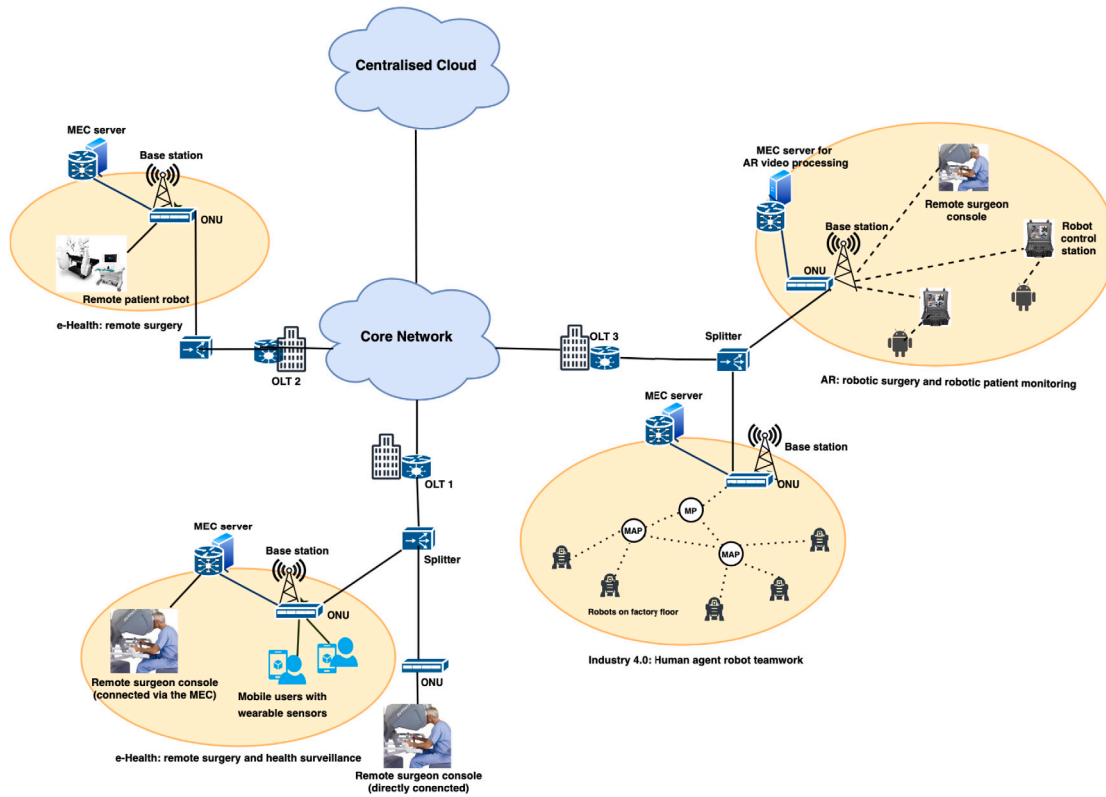


Fig. 3. PON-MEC use cases [57–59].

inset of Fig. 3 illustrates a possible configuration of HART. The MEC server in this case provides the required computation for ML and also for selecting agents/robots for different tasks. The applications of this nature specifically benefit from location and context-aware information available at the MEC.

3.2.4. Augmented Reality/Virtual Reality

Augmented Reality (AR), Virtual Reality (VR) and Extended Reality (XR) have become popular technologies in recent years to enhance the digital experience of users through providing a digital component to the real world, real life experience in the digital world, and a combination of both. This technology is currently being deployed in medical, educational, gaming, and streaming sectors [66]. For example, in healthcare, applications such as surgical training simulators and remote medical systems are now being explored using AR, VR, and XR [67].

In the work reported in [57], the authors discuss AR-based remote patient monitoring and robotic surgery to mitigate risk to medical personnel during the Covid-19 pandemic. AR-based remote monitoring is proposed to perform daily routines carried out by the medical professional using robots and is expected to have comparatively high precision than other conventional remote monitoring systems. AR-based robotic surgery is also proposed as means of getting more precision while performing invasive remote surgery, either when the surgeons are not available at the patient's medical facility or to reduce risk of exposure. The inset of Fig. 3 illustrates these two AR-based medical services implemented using PON-MEC integration. As emphasized in [57], improved precision is one of the main reasons to select AR as a favorable option for this application. To support this requirement, a large amount of video and audio data has to be processed. Further, life-critical applications of this nature require low latency and high reliability. These three requirements are well supported by the integrated PON-MEC architecture of Fig. 3.

The importance of integrated PON-MEC networks is clearly demonstrated through the use cases described above. More specifically, the MEC assists in offloading the computation and storage close to the end-user, thereby reducing the latency, while the PON facilitates connectivity through its high-capacity optical links. It is however, important to note that the integration of these two architectures does not guarantee that latency requirements will be met. For this purpose, end-to-end resource allocation across these heterogeneous networks has to be carefully managed through dynamic resource allocation algorithms, as will be discussed below.

4. Federated PON-MECs for H2M/R communications

4.1. Motivation

H2M/R applications such as tele-rehabilitation and cloud AR/VR play an important role in the aforementioned emerging e-health, industrial automation and AR/VR use cases [68]. Unlike conventional human-to-human (H2H) and M2M communications, H2M/R communication features real-time human control of and haptic feedback from distant machines/robots, thus allowing immersive human interaction with remote environments. However, the perception of real-time H2M/R communication requires millisecond round-trip latency between the human and machine/robot [4].

Recent literature in [4,69,70] suggests a converged network solution to meet this stringent round-trip latency. Fig. 4 shows such a network, comprising passive optical fiber and wireless access together with MEC functionalities. In conjunction with high-capacity PONs, MEC servers co-located at the COs and radio front-ends play the crucial role in bringing video, audio and proprioception resources closer to end-users. As importantly, through ML based traffic prediction, MEC can expedite haptic feedback delivery from the machine/robot to the human further improving latency performance and decoupling the human and machine/robot distance from the achievable end-to-end

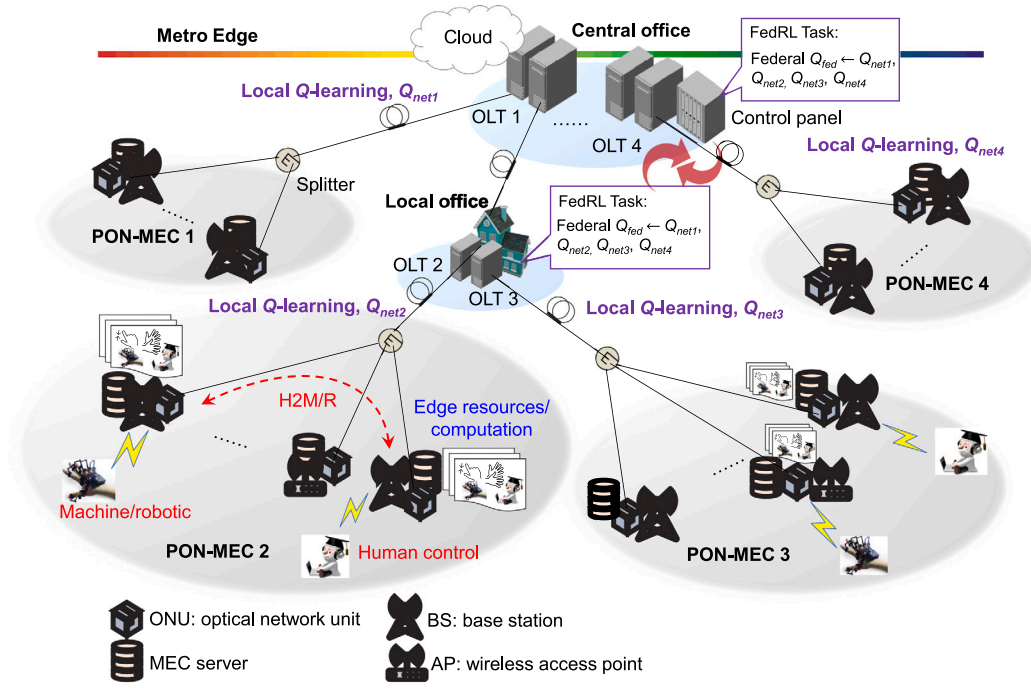


Fig. 4. FedRL framework over federated PON-MECs to support low-latency H2M/R communications.

latency [4,69,71]. This also helps reduce the amount of data traversing the network, thus relieving bandwidth to serve other latency-sensitive traffic [70,72–74].

4.2. Bandwidth allocation in PON-MECs

As H2M/R communications are highly interactive, the PON needs to support frequent data exchange amongst the MEC servers/ONU as shown by PON-MEC 2 of Fig. 4. As uplink bandwidth is shared, the allocation of uplink bandwidth to each MEC server/ONU to communicate with the CO, is particularly important to maximize bandwidth efficiency and latency performance. In that regard, a REPORT-GRANT process is adopted in classic dynamic bandwidth allocation algorithms to allocate bandwidth to ONUs [75]. Firstly, ONUs request for bandwidth BW_{REQ} based on their uplink buffer occupancy, in REPORT messages to the CO in a round robin fashion. Upon receiving a REPORT message, the CO then allocates bandwidth BW_{GRANT} to the ONU through the GRANT messages. The time interval between an ONU's consecutive uplink transmissions is known as a polling cycle, which in general is the scale of milliseconds [76]. The dominant latency is attributed to the waiting time a packet incurs while waiting to be transmitted upstream from the ONU. If an ONU is allocated insufficient BW_{GRANT} , its packets will spend an increased waiting time in the buffer, thus degrading the overall latency performance of the network.

In the baseline scheme, the CO grants BW_{GRANT} equivalent to that requested by an ONU but only up to a maximum, i.e. $BW_{GRANT} = \min\{BW_{REQ}, BW_{MAX}\}$. This scheme is considered a conservative scheme as any new packet arrivals will need to be queued and buffered until they are reported. On average, a packet will incur a latency equivalent to an approximate 1.5 polling-cycle time before being transmitted. To reduce this latency, a common solution is to predict the surplus bandwidth corresponding to new packet arrivals and allocate this bandwidth, BW_{BONUS} , along with BW_{REQ} [77]. Hence, with bandwidth prediction, the CO grants bandwidth BW_{GRANT} as follows:

$$BW_{GRANT} = \min\{BW_{REQ} + BW_{BONUS}, BW_{MAX}\}. \quad (1)$$

As can be gathered, BW_{BONUS} is a key decision parameter in which under-granting BW_{BONUS} can lead to increased waiting time while over-granting BW_{BONUS} can lead to bandwidth inefficiency. Ideally, BW_{BONUS} will reduce the waiting time of a newly arrived packet to 0.5 polling-cycle time. That is, newly arrived packets can be transmitted immediately without having to wait for the next polling cycle. To determine BW_{BONUS} , conventional dynamic bandwidth allocation algorithms exploit statistical prediction, i.e. using statistical features of the data received and transmitted in previous polling cycles. Different statistical prediction methods suit different data traffic profiles. For example, Bayesian estimation [78] and linear regression [79] are used to predict traffic with favorable time-average performance whereas moving average [80] is more suitable to predicting bursty traffic.

4.3. Reinforcement learning-based bandwidth allocation in federated PON-MECs

In recent years, supervised ML techniques have been investigated to improve the prediction accuracy of BW_{BONUS} [81–84] though the focus has not always been to meet stringent latency constraints. Specifically to support H2M communications over PON-MECs of Fig. 4, we proposed a federated reinforcement learning (FedRL) framework that leverages: (a) reinforcement learning (RL) to optimize BW_{BONUS} together; with (b) federated learning (FL) between multiple PON-MECs to minimize the time cost in executing sub-optimal decisions [85].

RL is renowned for optimizing decisions via exploration–exploitation and allowing the decisions to be self-adaptive to the environment [86]. In RL, an agent learns the best decision through strategic exploration of various decisions and receiving the rewards. As a trade-off, an optimal decision comes at the expense of time cost in executing sub-optimal decisions. In the short term, the decision exploration may negatively affect network performance, e.g. network latency.

To overcome the limitations of RL, we leverage FL to enable experience sharing between multiple PON-MECs that form a federation. By definition, FL is to enable distributed devices to collaboratively learn a shared ML model [87]. It is expected that all PON-MECs in a federation have similar states, e.g., with similar architecture and traffic environment, and thus sharing of decision experience between

the PON-MECs will accelerate individual decision making, eventually benefiting all PON-MECs in the federation.

In the subsections below, we first introduce local RL in individual PON-MECs, and then FL to expedite local learning in the proposed FedRL framework. Finally, we introduce the key network features to be considered in selecting federated PON-MECs for FL.

4.3.1. Q -learning in local PON-MEC

Firstly, RL is used in each local PON-MEC to make individual BW_{BONUS} decisions. With low latency as our goal, we define the immediate reward, termed R , of a BW_{BONUS} decision as the negative latency value when the CO grants a BW_{BONUS} to the ONUs. Specifically, as the ONUs transmit to the CO in a round-robin manner, in every T_{STEP} polling-cycle interval, an ONU evaluates the average latency d_{avg} of its packets and reports the reward $R = -d_{avg}$ to the CO. As such, during the t th step, the CO acquires a decision reward pair as $\{BW_{BONUS}(t), R(t)\}$.

Based on the decision reward pairs, the CO adjusts BW_{BONUS} for the next T_{STEP} in order to maximize the long-term expected reward. We adopt the Q -learning RL algorithm to measure the decision value and adjust decisions accordingly [86]. Further, we use a local decision-value function, denoted as Q_{net} , to indicate the expected reward of a decision in long-term. When the CO receives an $R(t)$, it updates Q_{net} , as follows:

$$Q_{net}(BW_{BONUS}(t)) \leftarrow (1 - \alpha)Q_{net}(BW_{BONUS}(t)) + \alpha(R(t) + \gamma(BW_{BONUS}(t+1))). \quad (2)$$

The $BW_{BONUS}(t)$ in (2) is chosen from a decision set D that comprises discretized bandwidth values. The parameters α and γ ($0 \leq \alpha, \gamma \leq 1$) control the learning rate and the weight on future rewards, respectively. With Q_{net} , CO determines $BW_{BONUS}(t+1)$ through a greedy policy given by:

$$BW_{BONUS}(t+1) = \arg \max_{BW_{BONUS} \in D} Q_{net}. \quad (3)$$

Eq. (3) follows the Bellman optimality equation and increases the convergence rate of Q_{net} and BW_{BONUS} . The process of (2) and (3) repeats until Q_{net} converges. Note that there are other policies in existing literature such as ϵ -greedy or softmax policy that have been used to explore decisions [86] but in general, these policies have a slower convergence time as compared to the greedy exploration. The granted bandwidth BW_{GRANT} to ONUs in (1) takes the updated BW_{BONUS} into account and is sent via GRANT message.

4.3.2. Experience sharing in federated PON-MECs

In FedRL, the use of FL to share Q experiences can help expedite local Q learning. Federated PON-MEC shares its $Q_{net,n}$ with other federated PON-MECs through updating a federal decision-value function, Q_{fed} , as illustrated in Fig. 4. Specifically, a heuristic rule is considered whereby when a $Q_{net}(BW_{BONUS}(t))$ is estimated, $Q_{fed}(BW_{BONUS}(t))$ immediately updates accordingly:

$$Q_{fed}(BW_{BONUS}(t)) \leftarrow Q_{net}(BW_{BONUS}(t)). \quad (4)$$

Then, each PON-MEC can refer to the shared Q_{fed} , instead of its local Q_{net} , to update its BW_{BONUS} decision using (3). Here, adjusting BW_{BONUS} decision based on the federated Q_{fed} is the major difference between FedRL and pure local learning without FL. In summary, the FedRL alternately exploits and updates a local Q_{net} and the federated Q_{fed} to adjust BW_{BONUS} decision in individual PON-MECs. The iterative learning and decision process involves the following key steps:

- Local BW_{BONUS} decision is adjusted based on the federated Q_{fed} in $BW_{BONUS} \leftarrow \arg \max_{BW_{BONUS} \in D} Q_{fed}$.
- Local Q_{net} is updated based on the reward received from executing BW_{BONUS} decision.
- Federated Q_{fed} is updated based on local Q_{net} updates. The above process repeats.

Algorithm 1: FedRL framework

```

1 Parameters:
2  $t$  — time stamp in unit of polling cycle
3  $BW_{BONUS,n}$  — decision in the  $n$ -th PON-MEC
4  $Q_{net,n}$  — local  $Q$  of the  $n$ -th PON-MEC
5  $Q_{fed}$  — shared  $Q$  by federated PON-MECs

6 Operation at  $n$ -th PON-MEC ONU:
7 if in transmission window then
8   Send buffered packets to the CO
9   if  $t = \text{polling cycle number}/T_{STEP}$  is integer then
10      $d_{avg} \leftarrow$  average uplink latency in  $T_{STEP}$  cycles
11      $R(t) \leftarrow -d_{avg}$ 
12   end
13    $BW_{REQ} \leftarrow$  remaining packets in the buffer
14   Send  $BW_{REQ,n}$  and  $R(t)$  (if  $R(t)$  is updated at  $t$ )
15 end

16 Operation at  $n$ -th PON-MEC CO:
17 if receive  $BW_{REQ,n}$  from ONU then
18   if  $R(t)$  updates then
19     Update  $Q_{net,n}(BW_{BONUS,k}(t))$  in (2)
20      $Q_{fed}(BW_{BONUS,n}(t)) \leftarrow Q_{net,n}(BW_{BONUS,n}(t))$  in (4)
21      $BW_{BONUS,n}(t+1) = \arg \max Q_{fed}$  in (3)
22   else
23      $BW_{BONUS,n}(t+1) = BW_{BONUS,n}(t)$ 
24   end
25   Grant  $BW_{GRANT,n}$  in (1) via GRANT message
26 end

```

The pseudo code of the proposed FedRL framework is described in Algorithm 1. Note that the operation of each PON-MEC is asynchronous to each other. Within an individual PON-MEC, ONUs report reward R to their CO following lines 9 to 12 and based on these reports, the CO updates its corresponding $Q_{net,n}$ as per line 19. Meanwhile, all federated COs maintain the federal Q function, Q_{fed} , as per line 20. This in turn allows each CO to make individual BW_{BONUS} decisions based on Q_{fed} as per line 21.

The FedRL framework has two favorable properties:

- Asynchronous learning: FedRL operates in a distributed fashion. The local Q -learning need not to be synchronized to enable FL in (4). In addition, one PON-MEC can immediately commence its local Q -learning from the latest Q_{fed} and greedily make BW_{BONUS} decisions using (3).
- Low complexity: FedRL operates in low computation complexity. The computation of Q value updates in (2) and (4) is in time complexity $O(1)$. The decision update in (3) is in a linear time to the number of tabulated BW_{BONUS} decisions.

Next, we elaborate on the conditions for experience sharing among PON-MECs in FedRL.

4.3.3. Key features for federated PON-MEC

In FedRL, the heuristic sharing in (4) is efficient when the federated PON-MECs are homogeneous. This is because FL typically expects homogeneous local learning environments for global performance and convergence guarantee [88]. When the local learning environments are heterogeneous, the FL may not converge to the correct decision needed in each local learning and the convergence rate may also be negatively affected [89].

In our context, the association between BW_{BONUS} decision in a PON-MEC and the corresponding latency performance is mainly related to the following network features based on studies in [81,90]:

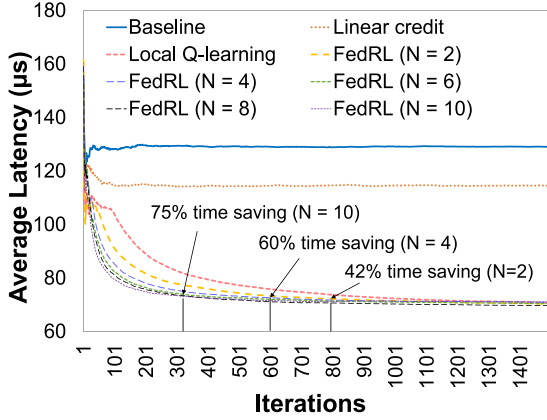


Fig. 5. Latency performance of tactile-haptic traffic (bursty traffic).

- number of ONUs;
- maximum OLT-ONU link distance;
- link data rate;
- traffic load; and
- packet statistics, i.e., length and arrival pattern.

In this paper, we consider PON-MEC networks having the same features as homogeneous and are suitable to share decision experiences in FL. Heterogeneous PON-MECs in general demand different BW_{BONUS} allocation in reducing latency. Hence, aggregating their Q values in FL may degrade the performance of local learning in individual networks. There are sophisticated methods such as model distillation and clustering to deal with FL in heterogeneous environments [89,91,92], which are beyond the scope of this paper.

4.4. Performance evaluation

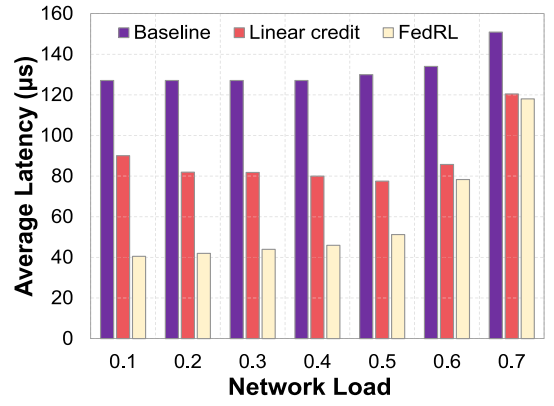
This section evaluates the performance of the proposed FedRL in reducing latency and expedited learning via FL. We also illustrate the FedRL performance considering different PON-MEC features.

4.4.1. FedRL performance

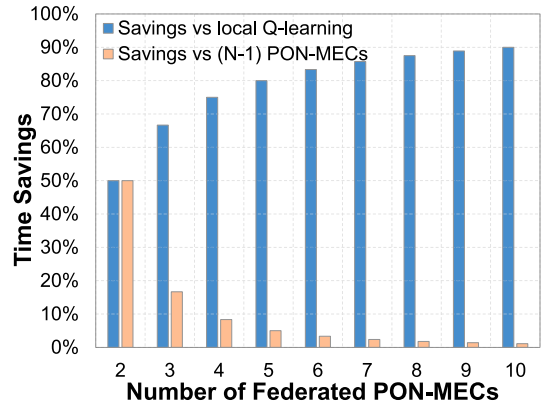
We perform simulations of the illustrative federated PON-MEC scenario shown in Fig. 4, whereby the maximum number N of federated 10G PON-MECs is 10. Further, each PON-MEC supports 16 ONUs and the CO and ONU distance is 10 km for illustrative purposes. The decision set D of BW_{BONUS} comprises 150 decisions ranging from 0 to 30 kB. The reward is reported every 10 polling cycles, i.e., one learning iteration. The time to process the REPORT message together with a reported reward $R(t)$, and to generate the GRANT message is 10 μ s [93]. The guard time between two adjacent ONUs' transmissions is 1 μ s.

We consider two types of traffic over the PON-MECs to validate the effectiveness of FedRL: (a) tactile-haptic traffic that are based on those collected from our H2M experiments; and (b) exponential user datagram protocol (UDP) traffic of a typical Internet environment [79]. Based on our experimental traces, the packet arrivals of the tactile-haptic traffic was modeled using a Generalized Pareto distribution with $\alpha = 0.9$ and $\gamma = 0.01$ values [69,85].

Fig. 5 shows the average uplink latency performance of tactile-haptic traffic. The latency performance of FedRL is compared with those obtained using the baseline (zero BW_{BONUS}) and linear credit (prediction via linear regression) schemes [79]. These two schemes result in a latency of more than 110 μ s while FedRL achieves a latency of ~ 70 μ s, a 40% reduction as compared to the baseline scheme. Moreover,



(a) latency performance

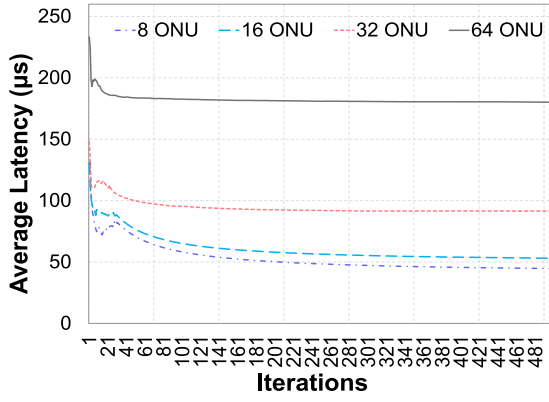


(b) learning convergence time

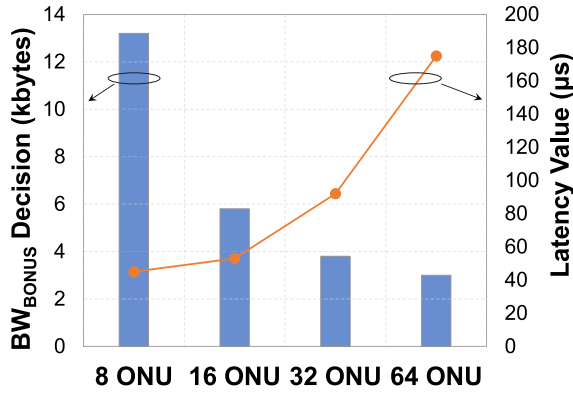
Fig. 6. Performance of FedRL with exponential UDP traffic.

results in Fig. 5 show that faster convergence is obtained with a higher the number of federated PON-MECs, N . Fig. 5 showcases the time-savings using FedRL to obtain an optimum BW_{BONUS} and stabilize latency performance as compared to when only using local Q -learning. Increasing the number of networks expedited learning. A federation of $N = 2$ PON-MECs saves above 40% time and a federation of $N = 4$ PON-MECs saves around 60% time for the average network latency to reduce to the minimum. With $N = 10$ PON-MECs, 75% time-savings is achieved.

With UDP traffic, Figs. 6(a) and 6(b) plot the average latency performance under different traffic loads ranging from 0.1–0.7 and the convergence time under different N , respectively. In Fig. 6(a), the latency performance of FedRL is shown to be superior to those of the baseline and linear credit schemes. With FedRL, bandwidth decisions are optimized by exploring and experiencing decisions over time. At low to medium traffic loads, the latency of FedRL is approximately 50 μ s, thus yielding a significant 60% improvement over the baseline latency. In Fig. 6(b), the blue column highlights the convergence time-savings of FedRL as compared to using local Q -learning in learning an optimal BW_{BONUS} decision. It can be observed that the higher the number of federated networks that share experiences, the higher the learning rate. A federation of $N = 5$ PON-MECs achieves a time-savings of 80%. When $N = 10$, the time savings is up to 90%. On the other hand, the marginal benefit of joining a federation diminishes as the number of PON-MECs increases. This can be observed through the orange columns representing the time-saving increments by adding a new PON-MEC to the federation. For $N > 5$, the time-savings is less than 5% as compared to a federation of $N - 1$ networks. Overall, Figs. 5 and 6 demonstrate that even at low N , federated PON-MECs sharing a Q_{fed} can achieve significant time-savings as compared to individual PON-MECs that use local $Q_{net,n}$.

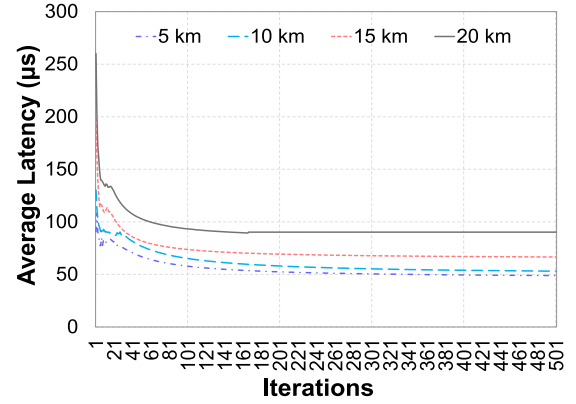


(a) latency performance

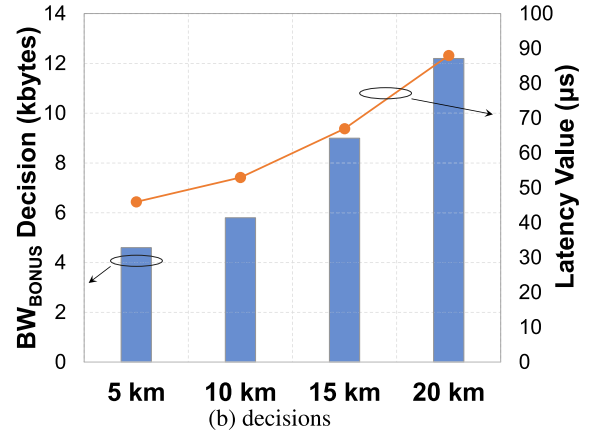


(b) decisions

Fig. 7. Performance with different ONU numbers (exponential traffic load 0.5 and $N = 4$ in FedRL).



(a) latency performance



(b) decisions

Fig. 8. Performance with different maximum link distances (exponential traffic load 0.5 and $N = 4$ in FedRL).

4.4.2. Performance under different PON-MEC features

To show how the features of PON-MEC impact the learning, Figs. 7 and 8 illustrate the latency performance and the BW_{BONUS} decisions learnt in FedRL considering learning environments with different numbers of ONUs and the maximum link distances, respectively. For illustrative purposes, we consider an exponential UDP traffic load of 0.5 and $N = 4$ PON-MECs.

Fig. 7(a) shows that when the number of ONUs increases, the final network latency increases. This is intuitive as more ONUs compete for bandwidth for uplink transmission will incur a longer packet waiting time at each ONU. Nonetheless, the convergence performance of FedRL is shown to be similar under different ONU numbers. In around 100 interactions, the average latency is lowered and stabilized in the presented cases in Fig. 7(a). Further, Fig. 7(b) details the final BW_{BONUS} decision learnt and allocated by the FedRL. First, the optimal BW_{BONUS} to allocate to the ONUs reduces when there are more ONUs in a PON-MEC. For example, in an 8-ONU PON-MEC, each ONU is granted 13 kB extra bandwidth on average to lower the latency to 45 μs. In comparison, in a 64-ONU PON-MEC, the BW_{BONUS} is only around 3 kB to the ONUs. The latency value corresponding to this BW_{BONUS} decision is 180 μs, which is significantly higher than the 8-ONU case.

Fig. 8 presents the impact of the maximum link distance in a PON-MEC on the BW_{BONUS} decision and latency performance. Similar to Figs. 7(a), 8(a) shows the dependency of network latency on link distance although the convergence performance of FedRL is minimally affected by it. In Fig. 8(b), we see that the BW_{BONUS} granted to ONUs increases as the maximum link distance in a PON-MEC is increased. This is because a longer distance causes a longer round-trip communication time between the CO and ONUs. Granting more bandwidth

allows more packets to avoid the round-trip report-grant process and thus reduces the latency. Results in Figs. 7 and 8 reveal that BW_{BONUS} decisions and their corresponding latency values are different depending on the network features of PON-MEC. For homogeneous PON-MECs considered in this work, the FedRL convergence rate primarily depends on how many PON-MECs to share their decision experiences instead of the detailed features of the federated PON-MECs.

5. Summary

The surge in latency-sensitive and computation-intensive applications due to the proliferation of 5G, Tactile Internet and H2M/R communication in conjunction with the exponential growth in connected devices, have pushed for architectural and capacity changes in both wired and wireless networks. To this end, we have discussed the evolution of high-capacity PONs and MEC, and as importantly how their inherent characteristics help in supporting latency-sensitive applications. Example use case scenarios that can benefit from integrated PON-MECs have also been detailed. As the integration of these two architectures does not guarantee that latency requirements will be met, end-to-end resource allocation across these heterogeneous networks has to be carefully managed through dynamic resource allocation algorithms. Specifically to support latency-sensitive H2M/R communications over PON-MECs, we proposed a framework based on federated reinforcement learning that harnesses reinforcement learning to optimize bandwidth allocation decision together with shared learning between multiple PON-MECs to maximize time-savings in reaching such decisions, further improving latency performance. Results show that the proposed framework achieves significant time-savings as compared to

the baseline and linear credit schemes. With the investigation of the framework for PON-MEC resource allocation showing overwhelming benefit in supporting latency-sensitive applications, our future research focus will be on the exploration of suitable machine learning techniques to accelerate reinforcement learning decisions within each PON-MEC and the identification of which specific PON-MECs to form a federation.

CRedit authorship contribution statement

Imali Dias: Writing – original draft, Investigation, Writing – review & editing. **Lihua Ruan:** Software, Experiment, Data collection, Validation, Writing – review & editing. **Chathurika Ranaweera:** Writing, Investigation, Writing – review & editing. **Elaine Wong:** Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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